SOFTWARE ENGINEERING
(LAB 1)

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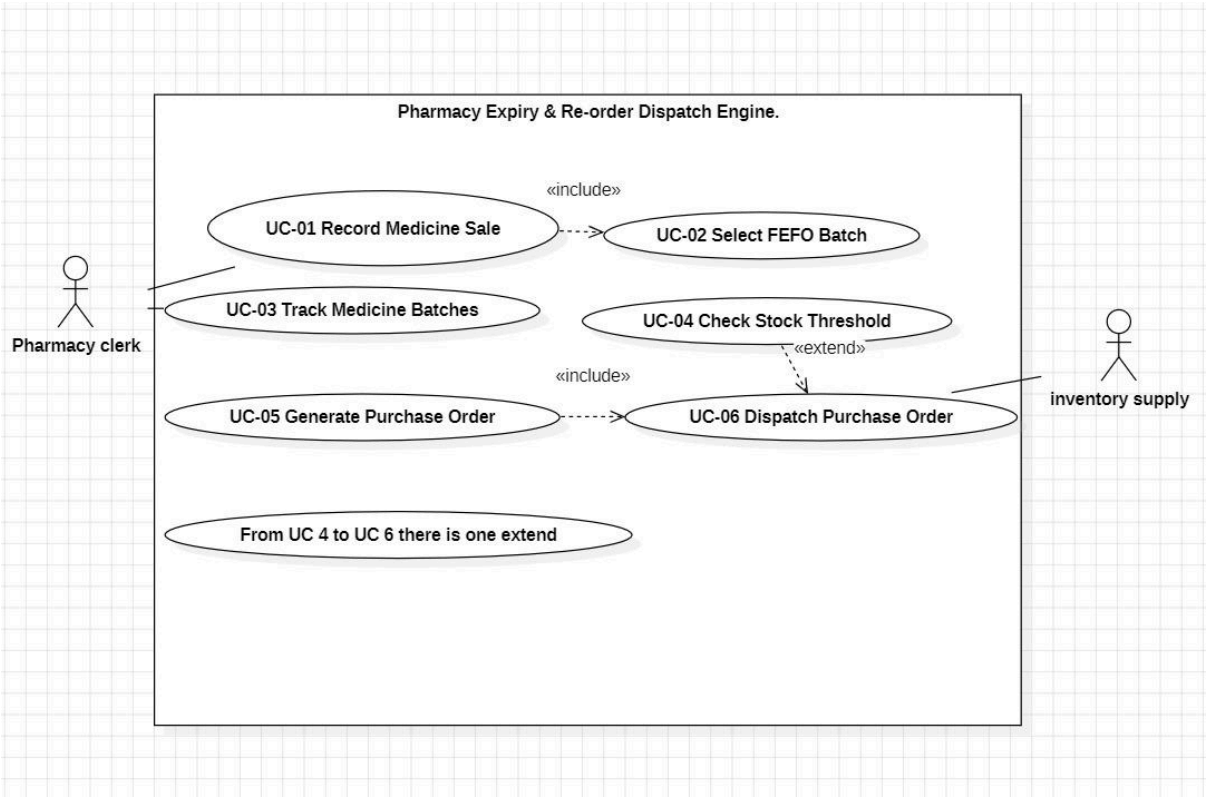
Pharmacy Expiry & Re-order Dispatch Engine

Requirements table:

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR - 001	Functional	The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during sales checkout	High	PASS: The oldest valid, unexpired batch is selected first. FAIL: A newer batch is selected while an older unexpired batch exists.	Ensures medicines closest to expiry are dispensed first and reduces expiry-related wastage
FR - 002	Functional	The system shall allow the Pharmacy Clerk to record medicine sales and update the available stock quantity for the selected batch.	High	PASS: After a sale is confirmed, the sold quantity is deducted from the selected batch and the updated quantity is displayed. FAIL: The stock quantity remains unchanged or is incorrectly updated	Keeps pharmacy inventory accurate after every medicine sale.
FR - 003	Functional	The system shall track and display the expiry date and available quantity of each medicine batch	High	PASS: The system displays the correct expiry date and current quantity for every registered batch. FAIL: Either value is missing or incorrect.	Allows the Pharmacy Clerk to identify valid batches and monitor inventory.
FR - 004	Functional	The system shall identify medicine items whose stock quantity falls below their configured safety threshold.	High	PASS: When stock becomes lower than the configured threshold, the item is flagged as below threshold. FAIL: An item below the threshold is not flagged.	Enables timely identification of medicines that need replenishment.
FR - 005	Functional	The system shall generate and dispatch a purchase order to the Inventory Supplier when an item's stock falls below its safety threshold.	High	PASS: A purchase order is generated and dispatched when stock falls below the threshold. FAIL: No order is generated or dispatched when the condition is met.	Automates replenishment and helps prevent medicine stock-outs.

NFR - 001	Non Functional	The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below the safety threshold.	High	PASS: During a simulated threshold-crossing event, the purchase order is generated and dispatched automatically without manual intervention. FAIL: Manual intervention is required or the order is not dispatched.	Supports automated replenishment when inventory reaches an unsafe level.
NFR - 002	Non Functional	The system shall update stock information and display the resulting inventory status within 2 seconds after a medicine sale is recorded.	Medium	PASS: At least 95 out of 100 simulated stock updates display the updated inventory status within 2 seconds. FAIL: Fewer than 95 updates meet the 2-second target.	Provides timely inventory information and supports efficient pharmacy operations

UML Use-Case Diagram:



Use-Case Flow Document:

Use Case: UC-01 – Record Medicine Sale

Primary Actor

Pharmacy Clerk

Preconditions

1. The Pharmacy Clerk is authorized to use the system.
2. Medicine batches and their expiry dates are recorded in the system.
3. At least one valid, unexpired batch exists for the requested medicine.

Postconditions

1. The medicine sale is recorded successfully.
2. The selected batch's stock quantity is reduced by the sold quantity.
3. The inventory status is updated with the new stock quantity.

Main Success Scenario

1. Pharmacy Clerk selects the medicine to be sold.
2. System retrieves all available batches of the selected medicine.
3. System checks the expiry dates of the available batches.
4. System identifies the valid batch with the nearest expiry date according to FEFO.
5. System selects that batch for dispensing.
6. Pharmacy Clerk confirms the sale quantity.
7. System deducts the sold quantity from the selected batch.
8. System updates the available stock quantity.
9. System checks whether the remaining stock has fallen below the safety threshold.
10. System records the completed sale and displays the updated inventory status.

Alternate Flow – No Valid Batch Available

4a. System determines that no unexpired batch is available.
4a1. System prevents the medicine from being dispensed.
4a2. System displays a message indicating that no valid batch is available.
4a3. Pharmacy Clerk is prompted to select another medicine or cancel the sale.
4a4. If the Pharmacy Clerk cancels the sale, the process ends without changing inventory.